

Lessons from Reality on Automated Container Terminals: What Can Be Expected from Future Technological Developments?

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Abstract

Automating container terminals is a consolidated trend in the port sector that is expected to grow. The main drivers for this growth are improving efficiency, reducing operating costs, remedying labor shortages, and increasing productivity. However, in practice there is no clear evidence confirming that automated terminals outperform conventional ones in handling productivity rates, cycle times, or vessel berthing times. Both the academic corpus and industry data show mixed results. Based on that, the goal of this article is to research productivity rates and factors affecting productivity, and understand to what extent future technological developments can enhance automated terminals. To do so, several sources of information are used and analyzed to shed light on the real productivity performance. The main conclusion gives no strong evidence linking automation with greater productivity, even though it might reduce terminal operators' costs. In fact, in many instances, productivity is worse. Moreover, currently, semi-automated terminals are still preferred to fully automated terminals because of better handling times. However, it is expected that future technological developments will likely overcome the current issues faced by automated terminals, when a technological paradigm shift takes places.

Keywords

marine, port, automation, container, terminals, policy and organization, disruptive technology, technology adoption

The world's first automated container terminal was ECT Delta in the Port of Rotterdam, the Netherlands, which started operations in 1993 (1). Its yard area and transfer from berth to yard area were designed without operators using automated stacking cranes (ASCs) and automated guided vehicles (AGVs). Over the years, the building of automated terminals has seen a consolidated trend, with European and Japanese ports taking the lead (2). As of 2023, there were 62 automated terminals (see Figure 1), and this trend is expected to grow exponentially (3, 4).

Nowadays, automation is already present in administrative or ancillary tasks, for example, terminal operating systems to control the movement and storage of containers, radio frequency identification, optical character recognition, or anti-sway systems in cranes (3).

Concerning automation of operations, the consensus is to use the semi-automated or fully automated classification. Semi-automated ports are those that have their

yard operations automated with ASCs or their family (automated rubber-tired gantry [ARTG], automated rail mounted gantry [ARMG], AutoStrad, etc.). Fully automated ports are those in which the yard and horizontal operations (from the berth to the yard area) are automated using AGVs. Figure 2 shows the schematic layout of a container terminal and the main automated equipment used.

However, a last type of automation has been recently tested, which is the remote control of ship-to-shore (STS) cranes. These still have to be operated by a dockworker, but they do it from a remote-control room. Examples are

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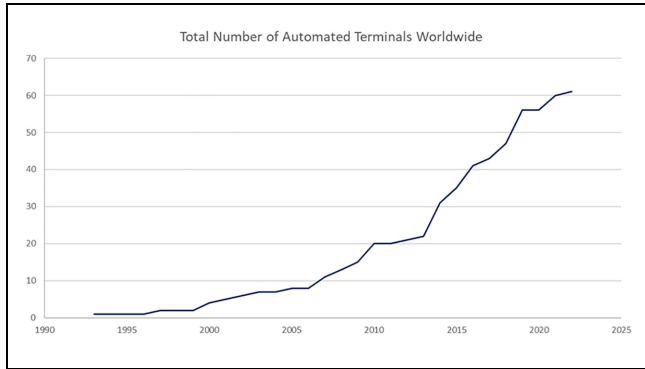


Figure 1. Total number of automated terminals worldwide.

found in Rotterdam (APMT at Maasvlakte 2, started in 2015), Shanghai (phase 4 of Yang Shan Terminal Complex), and Qingdao (Qingdao New Qianwan Container Terminal) (5).

Table 1, shows, to the best of the authors' knowledge, a complete list of the world's automated terminals as of 2023. There would be 62 terminals (63 with terminals to be built), being 4% in 2021 (or an estimated 5% in 2023) of the total number of terminals worldwide (5).

Drivers of Automation

Container ports are in a constant race to maintain their competitiveness, for example, the concentration of the shipping sector makes shipping lines more volatile (7). Consequently, to maintain their competitiveness, investments in new equipment or other activities (dredging) to improve cargo handling seem to be the only way to stand out.

Articulating competitiveness through automation is one way forward. Typically, three drivers are claimed:

reduction in operating cost (compared with manned operations), labor shortages, and higher productivity (8). As Huynh et al. state, automation is also the response to an intense competition between ports to become the fastest, cheapest, safest, and most reliable (9).

Concerning costs, it is evident that automating minimizes (but does not eliminate) labor costs. For instance, Donnelly found out that Qingdao (fully automated) labor costs had been reduced by 70%, according to the terminal's CEO (10). Despite this, it is acknowledged that there is high capital expenditure (4).

In relation to labor shortages, it is interesting to refer to Chen, who concluded—through stakeholder interviews—that Chinese ports lack dockworkers, as these jobs are not seen as attractive (between 25% and 40% abandoned training plans in Yangshan, Dalian, Tianjin, and Qingdao ports) (11). They alleged too low wages compared with manufacturing and food processing industries, and job hazards (200 people were injured during work at Yangshan Port in 2017). Consequently, this presents automation as a solution to keep competitiveness in Chinese ports.

Concerning productivity, there is a global need to reduce handling times. A combination of three factors might explain this. First, the use of bigger vessels (mega-vessel trend) potentially reaching 30,000 twenty-foot equivalent units (TEU) are increasing berthing times and slowing down handling unit productivity rates (12). Since mega-vessels mainly increase their beam, processing them can result in handling 80% more containers per meter of berth, comparing 15,000 TEU with 6,000 TEU (11).

In a similar vein, Mooney, using the Journal of Commerce's productivity database, shows an average drop in performance on ships larger than 14,000 TEU capacity (at 12 of the largest container ports) (13). Mooney suggests that the need for higher spreader hoists

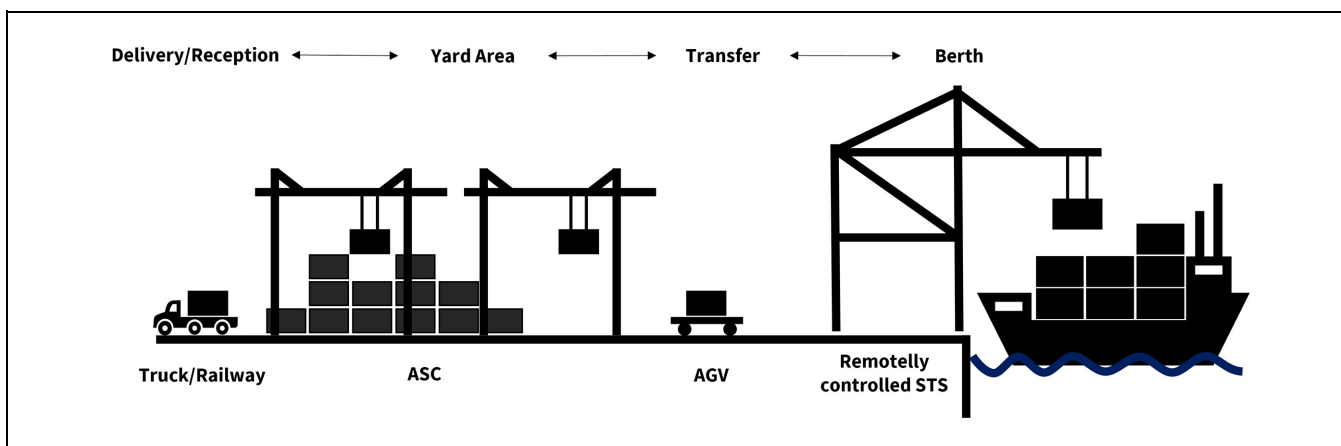


Figure 2. Container terminal layout areas and main equipment used.

Note: AGV = automated guided vehicle; ASC = automated stacking crane; STS = ship-to-shore.

Table 1. The World's Automated Container Terminals (Retrieved from Adam et al. and Knatz et al.) (3, 6)

Terminal	Port	First year of operation	Automation type	Technology
ECT Delta	Rotterdam	1993	Full	AGV, ARMG
Pasir Panjang 1-2-3 and 4-5-6.	Singapore	1997	Semi	ARMG
APMT-R	Rotterdam	2000	Full	AGV, ARMG
Thamesport	London	2000	Semi	ARMG
Altenwerder	Hamburg	2001	Full	AGV, ARMG
Fishermans Island	Brisbane	2002	Semi	ASC
Oi Container Terminal (Berth 6) Wai Hai	Tokyo	2003	Semi	ARMG
Evergreen Marine	Kaoshiung	2005	Semi	ARMG
DPW Gateway	Antwerp	2007	Semi	ARMG
Virginia International	Portsmouth	2007	Semi	ARMG
Korean Express Busan	Busan	2007	Semi	ARMG
Euromax	Rotterdam	2008	Full	AGV, ARMG
Tobishima Pier South	Nagoya	2008	Full	AGV, ARTG
Newport (Hanjin, HMM)	Busan	2009	Semi	ARMG
Newport (DPW)	Busan	2009	Semi	ARMG
Isla Verde	Algeciras	2010	Semi	ARMG
Taipei Port CT	Taipei	2010	Semi	ARMG
Kao Ming	Kaoshiung	2010	Semi	ARMG
Burdhardkai	Hamburg	2010	Semi	ARMG
BEST	Barcelona	2010	Semi	ARMG
Khalifa-TIL	Abu Dhabi	2012	Semi	ASC
London Gateway	London	2013	Full	AGV, ARMG
Global Terminal	New York and New Jersey	2014	Semi	ARMG
TraPac	Los Angeles	2014	Semi	ASC
SSA Manzanillo Int.	Colon	2014	Semi	ARMG
Yuan Hai	Xiamen	2014	Full	AGV, ARMG
DP World	Brisbane	2014	Semi	ARMG
HPH Brisbane	Brisbane	2014	Semi	ARMG
SICT-HPH	Sydney	2014	Semi	ARMG
Lamong Bay	Surabaya	2014	Semi	ARMG, ARTG
Jebel Ali 3	Dubai	2014	Semi	ARMG
APMT-MV2	Rotterdam	2015	Full	Lift AGV, ARMG
Rotterdam World Gateway	Rotterdam	2015	Full	AGV, ARMG
Patrick Stevedoring (AutoStrad)	Sydney	2015	Semi	ASC
PPT	Singapore	2015	Semi	ARMG
Middle Harbor	Long Beach	2016	Semi	ARMG
Tuxpan Port Terminal	Tuxpan	2016	Semi	ASC
Hanjin Incheon CT	Incheon	2016	Semi	ARMG
Liverpool 2	Liverpool	2016	Semi	ARMG
Victoria International CT	Melbourne	2016	Full	ASC, ARMG
Patikemas	East Java	2016	Semi	ARTG
Yangshan Phase 4	Shanghai	2017	Full	AGV, ARMG
Dublin Ferryport Terminal	Ireland	2017	Semi	ARTG
Hong Kong International Terminal	Hong Kong	2018	Semi	ARTG
Qianwai CT	Quingdao	2018	Full	AGV, ARMG
Khalifa Cosco	Abu Dhabi	2018	Semi	ARMG
DI, Laem Chabang	Thailand	2018	Semi	ARTG
Belwan International Container Terminal	Indonesia	2019	Semi	ARTG
Terminal C, Sohar	Oman	2019	Semi	ARTG
AMPT	Vado Ligure	2019	Semi	ARMG
Tanger Med 2	Tanger	2019	Semi	ARMG
Ferguson Terminal	Auckland	2019	Semi	ASC
Belfast Container Terminal	Belfast	2019	Semi	ARTG
Vizhinjam Port	India	2019	Semi	NA
Tianjin FICT	China	2019	Semi	ARMG
Red Sea Gateway Terminal	Saudi Arabia	2019	Semi	Remote STS
Norfolk International Terminal	Virginia	2021	Semi	ASC
Haifa Bay Terminal	Israel	2021	Fully	AGV, ASC
APM Terminal Pier 400	Los Angeles	2021	Semi	ASC
Hadarom Container Terminal	Israel	2021	Semi	ASC
Tianjin Second Port Container Terminal	Tianjin	2022	Fully	AGV, ASC
Nabeta (Nagoya)	Japan	2023	Semi	ARTG
Tuas Container Terminal Phase I	Singapore	2040	Full	NA

Note: AGV = automated guided vehicle; ARMG = automated rail mounted gantry; ARTG = automated rubber-tired gantry; ASC = automated stacking crane; STS = ship-to-shore; NA = not available.

and longer trolley distances is likely the cause of the drop of performance on vessels larger than 14,000 TEU, as crane cycles take longer on ships that have 19 to 22 containers across instead of 13 to 19 (13). Consequently, the loading and unloading efficiency of the port is under pressure.

Another important factor is that, as global trade volumes grow, many container terminals might be close to capacity—these are terminals that cannot grow on the surface. A port operating at or near its full capacity is highly vulnerable to the least amount of disruption, lacks the operating resilience to recover, and requires more unproductive moves (14).

Finally, another element to consider is flexibility in operations, compromised by growing trade volumes and increased volatility (high demand peaks induced by big vessels) (5). Uncertainty and volatility of traffic flows (big demand variation) suggest large variation of peak performances (flexibility) throughout the terminal (15). In contrast, terminals operated manually are more flexible and can be easily and rapidly adapted for new or different traffic (5).

It is often said that automation leads to higher productivity; however, when implemented in reality there is no clear or common ground around the topic. A large body of literature is dedicated to simulating or proposing, from a theoretical point of view, how productivity can be increased (16). However, concerning implementation, as Burgos et al. discover, the desired levels of automation are not always reached and are dependent on local conditions (17). In light of this, the goal of this paper is to give insight into and discuss common ground on productivity in automated container terminals, given the relevance it has on costs and terminal operators' managerial decisions. Moreover, by understanding the factors that limit productivity growth, one can discuss how future technological advances could potentially enhance automated terminals.

The remainder of the paper is organized as follows. The following section describes the methodological approach employed. The section after that assesses factors affecting productivity. The penultimate section dives into a benchmark of productivity rates. Finally, the Discussion section elaborates on the final conclusions concerning productivity in automated container terminals.

Methodology

The methodological approach used in this article has used four different sources of information to understand the current state of productivity at automated terminals, to a certain extent resembling an ethnographic study of a sector (container terminals):

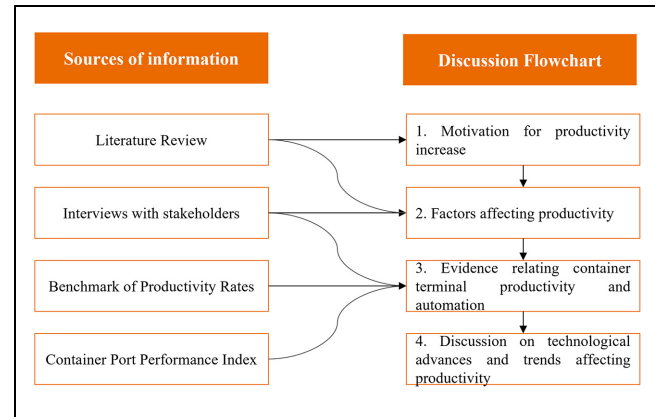


Figure 3. Article flowchart and sources of information used.

- 1) We carried out a literature review of scientific articles and a search through scholarly literature using research engines such as Elsevier Scopus and Google Scholar. Keywords included such terms as “automation,” “container terminal,” “ports,” “brownfield,” “productivity,” “efficiency,” and “frontier analysis.” The literature reviewed includes articles in scientific indexed journals and also industry reports from renowned institutions.
- 2) The outcomes of semi-structured interviews with stakeholders, technicians, and workers working in this field will point out some practical considerations and difficulties in automated terminals, that is, daily practicalities that do have an impact in the functioning of automated terminals but are not usually considered in the academic corpus.
- 3) The third element will be a benchmark of productivity rates. The sources for these rates are varied and include industry reports and news or port/container terminal statistics from terminal operators.
- 4) The fourth source will be the analysis of the World Bank Container Port Performance Index (WB CPPI). This is a crucial step as it is difficult to have access to private operator benchmarks. Finally, to find a relationship between automation and productivity we use the WB CPPI.

With these four sources of analysis, the intention is to describe the issue of automation at container terminals and productivity, and provide a detailed picture of the current situation. The structure of the discussion follows the reasoning below (illustrated in Figure 3). Firstly, the motivation for productivity increases at container terminals has been laid out above. Then, factors affecting productivity are set out to understand the extent to which automation can affect productivity at ports, by carrying

Table 2. Sources of Information Reviewed for the Discussion

Source of information	Analyzed elements	Elements consulted
Literature review	Scientific articles	45
Interviews with stakeholders	Stakeholders	11
Stated productivity results	Reports/news	21
Container port performance index	Terminals	50

out literature research. Thirdly, the evidence relating automated container terminals and productivity will be presented and compared against conventional or manned terminals by researching academic sources, reports, or news. Then, use will be made of the WB CPPI. Finally, the Discussion section will elaborate on the conclusions from the sources of information cited above. Table 2 summarizes the sources of information used.

Factors Affecting Terminal Productivity

The ease of use of key performance indicators might undermine the holistic understanding of the terminal and draw biased conclusions. The reason being that several factors influence the productivity of a container terminal, only some of which—that is, internal factors—can be controlled by terminal operators (14). Therefore, the goal of this section is to highlight which are the most common factors and understand to what extent a container terminal's productivity is subject to external factors or can be improved by automation.

Factors Identified in the Literature Review

External factors would be those that are not under the control of the terminal operator. These can be summarized as: trade volumes, shipping patterns, the ratio of import to export containers (which influences the number of empty containers handled at a terminal and the availability of container chassis), the size and type of ships accommodated by a terminal, and the landside capacities external trucks or rail area (18). Despite these factors affecting both conventional and automated terminals, the response to them can cause different outcomes in how the terminal works.

Concerning internal factors, container terminals differ considerably in size, function, and geometrical layout (in general under the control of the terminal operator), but they are principally made up of four subsystems: STS (berth line), transfer (horizontal transport), storage (yard area), and delivery/reception (transport from/to hinterland) (19). It is important to realize that a logistics

terminal is made up of a sequence of operations where the weakest link determines the overall productivity of the terminal (20).

Typically, the yard area or the transfer (between berth and yard area) are the weakest links. In that regard, Wilmsmeier et al. concluded that the most important issue is the coordination between vessel loading and unloading operations and the storage of containers in the yard, coordinating between subsystems (21, 22). Unsynchronized systems can either create congestion in the horizontal transport, or idle quay cranes or yard cranes. Many authors conclude that yard cranes are critical in the overall systems' productivity (23–25).

To contextualize this, Yang and Shen found that the most important yard facility performance factors were the number of horizontal vehicles, horizontal transport distance, yard crane handling efficiency, number of yard blocks, and speed of movement of yard cranes (26). Horizontal transport weight was 0.6 over 1, therefore having more influence, while efficiency and speed of yard cranes' weight was 0.24 (having less influence).

In relation to the berth line, Fourgeaud noted that quay cranes, one of the limiting phases of the handling cycle, is the time spent accurately positioning the spreader on a container (loading), or the container on a trailer (27). With advances in anti-sway devices, the problem is the capacity to deliver or remove containers from the quay without delaying STS operations.

Another important factor is the handling of exceptions. Containerized cargo popularity is because of the standardization and efficiency gains it brings to the shipping business. However, there are still parts in the handling of containers that veer off standardization. Such deviations increase the overall handling times, and Table 3 gathers such influencing factors. It is clear how any factor diverging from a standardized procedure, size, or shape, that implies an exception in the system, hampers handling productivity as pointed out by Chu et al. (28). Therefore, operational flexibility is found to be one important factor that hampers automated terminals' productivity.

Finally, Table 3 gathers the main influencing factors. These can be grouped into the following dimensions: shipping lines, vessel, container configuration/cargo composition, re-stowage, inter-terminal transfers, yard configuration, quay crane, and lashing.

Interviews with Stakeholders

The objective of interviewing stakeholders (terminal operators, technicians, technological providers, and dockworkers) was to understand the difficulties and practical constraints of automation experiences (gathered in Table 4). A total of 11 interviews were carried out

Table 3. Factors Influencing the Overall Container Terminal Productivity

Dimension	Factor	Reference
Shipping lines	Multi-port calls require re-stowage, that is, more crane movements compared with gateway ports. Transshipment or local landside distribution (the latter locates containers further in the yard) requiring longer intra-terminal transportation time.	Maritime Union of Australia (29)
Vessel	Lashing resources allocated affect the total berthing time. Because of safety requirements depending on the region, lashing cannot occur under a working crane, which means lashing commences after last container is loaded. Larger ships with wider beams and higher stacks require a longer time over the ship for cranes to pick containers. Larger ships generate peak demand, which can generate terminal congestion because of high resource allocation. Smaller ships result in more lateral movement of quay cranes, thus slowing down lifts/hour.	Maritime Union of Australia (29) Chen et al. (24)
Container configuration/cargo composition	The age of the ship and quality of lashing bars. Double-lashed containers. Mix of 20 ft and 40 ft containers or special containers (e.g., reefers) increase berthing time which require maneuvering or a change of spreaders.	Maritime Union of Australia (29) Fourceaud, and Esmer (27, 30)
Yard configuration	Spread-out terminals need longer horizontal transport. Stacked height containers mean more moves to access lower containers (higher handling costs).	Le-Griffin and Murphy (18)
Quay crane	Using large quay cranes for small feeder ships will be inefficient, as they are not adapted to handling smaller ships and will block slot space for the mega-ships. The number of cranes allocated (per berth) to a ship. Crane intensity may not lead to linear improvements in productivity.	International Transport Forum (5) Jordan (31)
Re-stowage	Lack of information about further transportation modes of the container (to plan the equipment needed for the container handling process). Most unproductive moves because of incomplete data. Also, changes of destination or schedule of vessels, resulting in sub-optimal location of already stored containers.	Mazloui and van Hassel (32) and Voß et al. (33)
Horizontal transport	While an increase in the number of AGVs leads to a decrease in the average ship service time, using many AGVs is very often not an efficient way of shortening the ship service time, in fact it might cause congestion in the network, and thus a decrease in the terminal's capacity. Using more STS cranes implies higher obstacles for AGVs, which can cause congestion and slowdown as they require clear space to handle the containers. Congestion (e.g., Rotterdam and Antwerp) being created from the off-loading process of mega-vessels because of the bottleneck moving containers out of the port as it is unloading or loading a vessel (inefficient downstream chain). Remote control operators have difficulties gauging the effect of wind on the container or weight feeling, for example, if a twist lock has not been properly disengaged. Full automation (automated guided vehicles and automated stacking cranes) is not easy to achieve because of the presence of external trucks. Synchronization between subsystems: quay, horizontal transport, and yard area. If vehicles do not arrive on time, quay cranes or yard cranes might be delayed, or congestion can be generated.	Pjevcevic et al. (34) Maritime Union of Australia (29) Chu et al. (28) Maritime Union of Australia (29) Achterkamp (15) Lee et al. and Luo and Yue (35, 36)

Note: AGV = automated guided vehicle; STS = ship-to-shore.

Table 4. Practical Constraints Raised by Technicians, Technological Providers, and Dockworkers during Interviews

Group	Automation area	Keywords	Concern
Technician/ chief of IT	Remote control	Technological maturity	Remote control of ship-to-shore could be done because they have telecoms already installed (optic fiber). It is currently not done because there is a physical need for operators. Operators on top of the crane have more flexibility and operate faster.
Technological provider	General	Technological maturity	Currently, most advanced sensors need to be adapted to the harsh marine environment. Despite advances, the end-technology is still immature and will have to be adapted if new, more reliable communication technologies are deployed (e.g., 5G).
Technological provider/ chief of IT	Horizontal transport	Technological maturity	Automated horizontal transport needs very low latency to avoid slow response time. On the current communication networks, only a few automated guided vehicles can be connected without loss of bandwidth and low latency. This is expected to be improved with next generation mobile networks (e.g., 5G).
Former director of port	General	Feasibility	Automation cannot be applied in all ports: each port facility is different, and it is a very wide world and activities are very different. There is no perfect solution that can be considered to be optimum.
Dockworkers	Yard area	Technological maturity	Some real problems are that automated stacking cranes (ASCs) sometimes miss the pins and they lift the entire truck up to the stacking height.
Dockworkers	Yard area	Hinterland connection/delivery reception	The problem is also when there is a lack of lorry drivers, for example on Fridays or weekends or at nights. When there is a congestion problem, the planning is rescheduled, but it may happen that piled-up containers aggravate the problem by making ASCs inefficient.
Dockworkers	Yard area, horizontal transport	Technological maturity	Every time there is an exception/unexpected situation, an operator takes control of the crane. Over time, there are more exceptions because the system is more complex. Moreover, with time, physical parts move, for example, cranes and sensors, and therefore optical and laser sensors de-calibrate or are worn out.
Dockworkers	Remote control	Technological maturity	Remote operators miss the context. For example, often a container is locked, and they feel resistance, and therefore immediately correct it. However, a remote controller does not feel this.
Dockworkers	Berth	Inequalities	To make up for larger ships (mega-vessels) taking more time to load/unload, more quay cranes are allocated. Consequently, mega-vessels cause smaller ships to be delayed, generating inequalities and increasing smaller ships' costs.
Dockworkers	General	Technological maturity	There are also collisions between cranes and automated straddle carriers.
Dockworkers	General	Technological maturity	There are still collisions or accidents. This coexists with the reality that there is still the need to use lashers and checkers on board ships.

during a span of 5 months (2021–2022) in an open format. Stakeholders were located in Australia, Singapore, United Kingdom, Spain, South America, and North America. Despite it being difficult to get responses from terminal operators on automation, the interviews which were eventually carried out bring value to the discussion. For privacy reasons no further details are disclosed.

The main common thread observed is technological maturity. When an exception occurs, a dockworker has to take over control, therefore increasing handling times. As they suggest, there are still incidences or problems with the precision of the automated technology, as well as lack of flexibility. Moreover, they suggest that current technology does not allow for further widespread automation (e.g., AGVs), as a minimum requirement of bandwidth and low latency are needed. Also, external factors, such as the lack of trucks for delivery, slow down the operations inside the terminal.

Benchmark of Productivity Rates

Benchmark Context of Conventional Terminals

This section puts together several references to showcase certain stated productivity results in automated terminals. Before that, it is important to distinguish what productivity indicators are available. First, quay crane or STS unit rates, since they are not typically the bottleneck, can be taken as a measure (proxy) of the whole terminal's productivity. Besides, berth line productivity factors in the number of quay cranes allocated and are a good indicator of the total berthing time of a vessel. However, there may be other managerial decisions, or external factors, influencing the allocation of cranes at ships, therefore influencing this indicator.

Concerning quay crane unit productivity, Chu et al. state that manned operations can consistently reach 30 containers/hour (28). Humphreys et al. stated typical gross moves per quay crane between 25 and 30, in 2017 (37). GHD said that the typical upper-end productivity for a single lift crane is recognized as 35 moves/hour (38). They also reported that the world record for a single lift crane is 75 moves/hour.

With respect to berth line productivity, Humphreys et al. foreshadowed that the mega-vessel trend would need at least 200 moves/hour at the berth line, or 300 according to Achterkamp for mega-hub terminals (15, 37). To achieve a berth performance of 300, the target STS productivity should be 50 moves/hour on average.

Analyzing 57 large ports, in 2006, Cullinane et al. calculated those ports above 1 million TEU/year presented decreasing returns-to-scale, and the other way around (7). This means that unit productivity is not enhanced in larger ports when increasing their input quantities. These findings could be explained by the discrete investment

nature of terminals, for example, indivisible port investment, the consequent commercial risks involved, and the level of competition in the market. On the other hand, economies of scale were present in all terminals (decreasing unit costs).

Rates at Automated Ports

Xu and Ishiguro conclude that, from 24 Asian terminals, the automated container terminals ranked unsatisfactory, and only one of them ranked slightly above the average efficiency values of the sample (39). At the very least, there is no reason to believe that automation is not a distinguishing factor in overall terminal efficiency.

In a similar vein, Burgos et al. analyzed automation of brownfield container terminals (17). On completion of automation projects, there were both increases and decreases. They stated that ARTGs presented slower cycle times, especially in the case of fully automated terminals where achieved productivity is lower than manned ones. Therefore, changes in productivity cannot be only attributed to automation, and will depend on the terminals' conditions and the solution adopted. Moreover, they conclude that 79% of the sample chose semi-automated terminals. The respondents alleged fewer labor problems, improved vessel productivity, less required space, better agility in dealing with disruptions, and lack of maturity in the control systems for full automation.

Another approach has been to analyze samples from terminals using non-parametric production methods (data envelopment analysis or stochastic frontier methods). Their common conclusions are that there is no reason to believe that automated terminals rank higher than conventional ones.

On the contrary, by analyzing several major factors affecting the operational performance of Kaoshiung's Evergreen Terminal, Yang and Shen found that the combination of automated container terminals and multimodal transport can make port operations better (26). When assessing particular crane yard equipment, ARMG is recognized as yielding the best operating performance, and is followed by ASC, RMG, SC, and RTG. Automated equipment outperformed conventional equipment between 5.5% (compared with RMG) and 50% (compared with RTG).

While these results seem to better contextualize the state of automated container terminals, it is also true that, over time, these have been enhanced. According to Achterkamp the introduction of the second trolley with lashing platform at CTA Hamburg made it possible to decouple the manual handling of twistlocks and give more flexibility to the AGV on the quay, allegedly improving productivity from 25–30 to 30–35 quay crane moves/hour (or 150 berth moves/hour in Rotterdam)

(15). The prospect is to reach 45–50 moves/hour (or 300 berth moves/hour) using technical solutions such as multi-lifting, dual cycling, and decoupling of the twist lock handling (15). This highlights the potential of alternatives, rather than automated solutions.

Despite automation, there are other physical limitations that limit the growth in productivity. Blocks with 2 ASCs per block show that the ASC system is spending 40%–50% of its moves on housekeeping even during peak demand. In principle, with perpendicular ASC blocks, complicated stacking algorithms and conflicts of priority cannot be avoided. Adding more blocks for serving megavessels conflicts with space limitations at ports (15).

Productivity Benchmark Summary

As a summary, Table 5 shows a list of productivity rates researched. As can be seen, most reported rates are between 25 and 33 moves/hour per berth crane. Except for Barcelona BEST (semi-automated) reaching 40 moves/hour and Qingdao (fully automated and remotely controlled STS cranes) reaching an average of 36 moves/hour and a peak of 56 moves/hour.

World Bank Container Port Performance Index (WB CPPI)

The last methodological approach is to examine the WB CPPI which was launched in 2019, aiming to objectively benchmark productivity between ports in a fair manner, resting on total port time (37). The relationship between automated and conventional terminals and said index can be found in Figure 4.

Figure 4 shows that there is no clear clustering of automated versus conventional container terminals in relation to their performance index or their annual throughput. First, it supports other articles' inferences that automated terminals are not restricted to large throughput volumes. Second, automated terminals are scattered throughout all levels of the productivity index, mixed between conventional terminal performance. This is supported by the DEA (Data Envelopment Analysis) analysis carried out by Wiśnicki et al.; they did not find a relationship between terminal efficiency and terminal automation level, so conventional terminals were found to be at the same level of productivity as automated ones, or vice versa (51). There are other context-specific factors that affect efficiency, as well as other cited works (8, 17, 39).

Discussion

Given the mixed evidence found on productivity, fully automated container terminals are questionable in this regard. In practice, their implementation is a result of

other drivers (e.g., labor cost reduction). The discussion aims to evaluate what the challenges are and why productivity is hampered compared with manned operations and to what extent future technological developments can enhance automation.

The different sources of information researched are aligned in that, with respect to productivity alone, automation at container terminals is not improved (compared with conventional ones). The reasons as to the difficulty in raising productivity are diverse, but here are grouped into external and internal factors.

Concerning external factors, these are mainly the high peak demands and high volumes of trade. These can create congestion in the terminal and lead to sub-optimal stowage or storing plans, or slowed down operations because of the terminal being close to capacity. It is important to acknowledge that these factors are not controlled by the terminal operator. However, equipment or operations could be adapted or optimized to face this situation.

Internal factors, or those that can be addressed by technology, are as follows. First, there are exceptions, unexpected events, or lack of flexibility. These, on many occasions, slow down operations because the system is not capable of rectifying a behavior or adapting to the external conditions in real time. Also, there are technical aspects such as decalibration or the maturity of the sensors and end-technology. A system dependent on a plethora of devices is vulnerable to the disruption or malfunctioning of any of its parts.

This can lead to workers having to take over in unexpected events. Related to that, Burgos et al. raise the issue of vulnerability (17). The main drawbacks of automation were given to be unpredictable technical problems/high impact of failures. Respondents recommended planning fallback solutions and building redundancy and resiliency into all systems. Of course, this approach compromises the cost balance if too much operational redundancy is needed. In short, the current maturity of end-technology is still far from providing a reliable and flexible system.

In line with technological maturity, interviews with stakeholders highlighted that the current telecommunication technology (on which end-technology is mounted) allows for only a certain number of devices to be connected at the same time to ensure bandwidth and low latency. Low latency is key for automated vehicles (in particular AGVs) to be able to respond quickly in case of an unexpected event or close collision. That is also a limiting factor for the advancement of automated terminals that could potentially be solved with next generation communication technologies. To address this, newer telecommunication technologies, for example 5G, could provide more robust platforms to ensure such industrial applications.

Table 5. Benchmark on Productivity Rates

Source	Terminal	Method	Automation analyzed	Main conclusion
Burgos et al. (17)	Brownfield terminals: Australia, Germany, Dublin, Japan, Netherlands, ME, Panama, Turkey, UK, Virginia	Operator questionnaire	Full-automated and semi-automated terminals	Brownfield terminals reported slower ARTG cycle times.
Yang and Shen (26)	Port of Kaohsiung	Grey relational analysis based on questionnaires	Container yard area	Productivity results were mixed. ASCs and ARMG outperform conventional straddle carriers and RTGs by 10% at most.
Kugler et al. (40)	na	Interviews	Automatic twist locks	Technological barriers to its implementation
Tanger Med Port Authority (41)	Tanger Med, container terminal 4, 2021	Reported rate	NA	32 moves/hour per STS crane
Safety4Sea (42)	Algeciras, March 2019	Reported rate	NA	33 moves/hour per STS crane, 9 cranes working
Camm (43)	ECT Delta	Reported rate	Fully automated terminal	28 moves/hour per crane
Österreichische Verkehrszeitung (44)	Euromax, Rotterdam, 2017	Reported rate	Fully automated terminal	200 berth moves/hour or 28 moves/hour per STS crane
Wee (45)	BEST, Barcelona, 2018	Reported rate	Semi-automated terminal	40 moves/hour per STS crane
Raimund (46)	Gioia Tauro, 2020	Reported rate	NA	25 moves/hour per STS crane
Maritime Union of Australia (29)	Australia, 2021	Reported rate	ARMG	Manual interventions every 4th lift
Chu et al. (28)	2018	Questionnaire	NA	showcasing slow ramp-up period
Maritime Union of Australia (29)	DP World Brisbane	Reported rate	ASC	Realized productivity improvements decreased 11% with respect to the previous situation.
Maritime Union of Australia (29)	APMT-Maasvlakte 2	Reported rate	Remotely controlled STS	27 GMPH in 2011 with conventional equipment. In 2021, 26.2 GMPH with ASC.
Aldama (47)	Qingdao, 2017-2019.	Reported rate	Full automation and remotely controlled STS	Average of 25 moves/hour per STS crane
Donelly (10)	Qingdao, 2022		Full automation and remotely controlled STS	2017: 26.1 moves/hour 2018: 33.1 moves/hour 2019: 35 moves/hour (per STS crane)
WorldCargo News (48)	Pakistan Hutchinson Ports	Interview with Uno Bryfors, senior vice president of ABB	Remotely controlled STS	36.2 moves/hour per STS crane on average. Setting a peak of 52.76 moves/hour.
Cuppens (49)	DP World terminal	Reported rates	Remotely controlled STS	After more than 500 training hours, trainees reached between 23 and 29 moves/hour per STS crane.
Hadland and Tomar (50)	Antwerp, Bremerhaven and Hamburg	Statistics	NA	Slow ramp-up period. After 5 months, 5 moves/hour slower than manned operations. After 21 months, remote STS cranes reached 30 moves/hour per crane. Berth productivity from 119 to 90 moves/hour, from 2019 to 2022

Note: ARMG = automated rail mounted gantry; ARTG = automated rubber-tired gantry; ASC = automated stacking crane; GMPH = gross moves per hour; RTG = rubber-tired gantry; STS = ship-to-shore; na = not applicable; NA = not available.

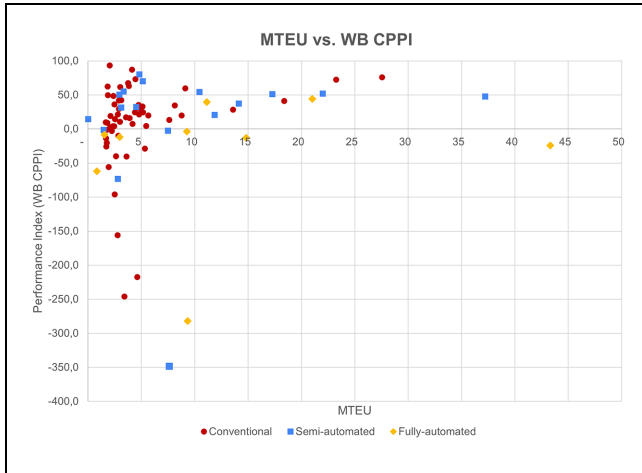


Figure 4. Million twenty-foot equivalent units (TEUs) (x-axis), World Bank Container Port Performance Index (WB CPPI) (y-axis), and clustering of semi-automated, fully automated, and conventional container terminals.

Another internal factor limiting productivity is the container stacking problem and the lack of information that generate re-stowage and unproductive moves (32). As Voß et al. comment, the optimization problem is complex, as a container terminal is a dynamic system (33). The terminal consists of various dynamic and continuous interactions inside the stacking area, and other elements, including ships, cranes, SCs (Stacking Cranes), AGVs, or trucks and next container destinations. These problems of course affect both automated and conventional terminals. Chu et al. identified lack of quality and structured data as an issue that makes it difficult to monitor operations and allow automation (28). To address this, the growing trend of digitalization, monitoring, and technologies such as the Internet of Things (IoT) or Blockchain are needed to solve the container stacking problem.

Finally, overall productivity is affected by other factors, such as lashing, that still has to be done manually. For example, as Kugler et al. studied, it is not clear whether automating the twistlock handling could yield faster handling rates; the main barriers are currently technological (40). In view of all these examples, container operations still need the flexibility manned operations have.

Therefore, all the evidence and issues discussed suggest that the main technological challenges limiting the improvement of productivity at automated container terminals can be grouped in the maturity of the end-technology and computational readiness. Consequently, there seems to be no reason to believe that, over time, this can be improved with technologies that, given their capabilities, are expected to revolutionize the industry sector (IoT, 5G, machine learning, or artificial intelligence [AI]), an outlier example of which is the fully

automated container terminal of Qingdao, reaching more than 40 quay crane moves/hour.

Consequently, it seems reasonable to conclude that, nowadays, automation in container terminals is motivated by other drivers such as efficiency (in reduction of unit costs) or other intangible factors (labor shortage, local conditions, dealing with union workers, etc.). Moreover, there might be alternative drivers, as Yang and She discussed, such as the electrification of equipment (ASCs come with electrified propulsion, in comparison with traditional diesel-fueled straddle carriers, reach stackers or others), which could be done for manned operations, but the industry seems to be electrifying only automated equipment (26). Despite being far from the automation discussion, decarbonization is expected to be another shifting force.

On determining which terminals are automated, some authors estimate the feasibility of automation for over 500,000 TEU/year (52). However, as Garrido Salsas et al. found out, only 11% of the fully automated terminals and 22% of the semi-automated terminals handle between 250,000 and 1 million TEU, meaning that they are not exclusive of the largest terminals (4). They suggest that automation occurs at terminals that target 10,000 TEU vessel calls. They also found that ROI (Return on Investment) was realized after between 5 and 6 years for 29% of the terminals and 61% estimated it would take more than 6 years (from a sample of 32), which is in line with the evidence found using the WB CPPI.

Also, as Burgos et al. discuss, most existing automated container terminals were greenfield developments, for which automation is already the norm, yet brownfield automation is gaining momentum and is expected to become more relevant in the coming years (17).

Considering the above, it seems that the sector is embarking on automating facilities because of other drivers the benefits of which compensate for the loss of productivity. As these early adopters are a testbed for automation it points to enhancement of the current technology, especially with the development of other technologies such as the deployment of new mobile network technologies (e.g., 5G) that allow the widespread use of IoT or ensure reliable operations without the loss of telecommunication performance. This could allow the development of digitalized terminal models, that would tackle the dynamic and online optimization methods (for storage planning) (53). Another driver of automation is that the computational burden could be relieved as next generation mobile technologies (e.g., 5G) allow real-time data collection and edge computing, and automatization.

In fact, the implementation of 5G technology is a real disruption for the performance of automated and digitalized ports, given its features such as bandwidth or low

latency and, in particular, its reliability (in contrast with other telecommunication technologies, e.g., 4G), most importantly because they address the main issue of technological maturity of the base telecommunication network (54). This points to the current technological state of the practice being on the verge of overcoming the main limitation of why productivity (or efficiency) at automated ports is limited. However, there are still other factors that will come into play with the connectivity of ports that will modulate or change the landscape of automated ports, such as its new forms of vulnerability (cybersecurity or dependence on telecommunication technologies) or its new relationship with labor, or even the market power change it might bring about.

Finally, the deployment of IoT would lead to a growth of Big Data collection and the enhancement of AI. This is particularly important in relation to the lack of flexibility issue. A priori, the handling of unexpected events and exceptions, or contextual adaptation, is only possible with the application of AI (55).

In view of the issues and technologies suggested above, it is important to highlight the differences between the potential effects of digitalization and automation. In short, digitalization could help overcome the container stacking problem, foresight cargo flows, or aid workers to optimize the current paradigm in container terminals. On the other hand, fully automated container terminals cannot be conceived apart from as a future scenario which changes the current paradigm. To arrive at that scenario, technology still needs to prove it can give operational reliability and flexibility, especially in a very complex and dynamic system as is the port. That could be achievable when there is a leap in the capabilities of technology, which start to show early progress in other sectors (computer vision, natural language, or edge computing) (56–58).

In a nutshell, it is expected that automation will increase exponentially once digitalization of ports starts to be popularized and end-technology evolves, induced by the deployment of new telecommunication technologies that already provide the characteristics to allow for reliable and robust industrial use-cases need. However, the port ecosystem needs to be aware that, with the increase in connectivity, the port will become more vulnerable to any technological disruption.

Conclusions

The main goal of this work was to review the automated container terminals and provide a common ground for both academia and industry. The evidence suggests that automated terminals are currently not much more productive than conventional ones. For example, as regards berth cranes, automated terminals reach peak performance between 25 and 33 gross moves per hour

(GMPH), while conventional ones consistently reach around 35 GMPH. Despite this, automation is progressively gaining market share (in 2023 reaching 62 semi- and fully automated container terminals worldwide) as it may be feasible from an economic point of view.

In a nutshell, the reasons for productivity being behind are mainly because of technological maturity. As it has been further argued in the Discussion section, device flexibility and adaptability to the environment or handling of unexpected events is not readily available. Examples of this are operators taking over ASCs in unexpected situations, AGVs not having latency low enough, or how remote operation of quay cranes (or yard cranes) miss the context to quickly react to deviations in operation, which slows down the whole container terminal system. Therefore, advancing current automated terminals is subject to technological development, and a potential example is found in the outlier fully automated terminal of Qingdao (with stated average 36 quay cranes GMPH in 2022).

The effect of technological developments on the current barriers for automation at ports should be distinguished in two dimensions. Progress in digitalization at ports is expected to improve or optimize while maintaining the current technological paradigm, aiding workers in a scenario where technology is a complement. However, to be able to see fully automated terminals, a paradigm shift in technology is required.

The reliability and operational flexibility needed to maintain cargo flows in a complex, dynamic, and very demanding environment such as a port requires not a progressive improvement of the current scenario but a comprehensive disruption in the technological capabilities, for which the synergies stemming from the development of different current fields (5G, IoT, Big Data or AI) might be the first step. Bearing that in mind, further research is needed to distinguish how progressive stages of technological development may affect automation, especially on different types of terminals and different container terminals' functional units, in relation to the current barriers.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: G. Majoral, A. Reyes, S. Saurí; data collection: G. Majoral, A. Reyes, S. Saurí; analysis and interpretation of results: G. Majoral, A. Reyes, S. Saurí; draft manuscript preparation: G. Majoral, S. Saurí. All authors reviewed the results and approved the final version of the manuscript.

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References

1. Evers, J. J., and S. A. Koppers. Automated Guided Vehicle Traffic Control at a Container Terminal. *Transportation Research Part A: Policy and Practice*, Vol. 30, No. 1, 1996, pp. 21–34.
2. Steenken, D., S. Voß, and R. Stahlbock. Container Terminal Operation and Operations Research – A Classification and Literature Review. In *Container Terminals and Automated Transport Systems* (H.-O. Günther, and K. H. Kim, eds.), Springer, Berlin, 2005, pp. 3–49.
3. Knatz, G., T. Notteboom, and A. Pallis. Container Terminal Automation: Revealing Distinctive Terminal Characteristics and Operating Parameters. *Maritime Economics and Logistics*, Vol. 24, 2022, pp. 537–565. <https://doi.org/10.1057/s41278-022-00240-y>.
4. Garrido Salsas, J., S. Saurí, S. E. Raventós, C. Rúa, and J. Torrent. Emerging Trends Defining the Future Role of Ports: Application of the Delphi Method. *Transportation Research Record: Journal of the Transportation Research Board*, 2022. 2676: 571–585.
5. International Transport Forum. *Container Port Automation: Impacts and Implications*. International Transport Forum Policy Papers, No. 96. OECD Publishing, Paris, 2021.
6. Adam, V., C. Comtois, and B. Slack, B. (2021). *Innovations et transformation de la main-d'œuvre maritime et portuaire: occasions et défis d'adaptation au Québec*. CIRRELET-2021-50. Bureau de Montreal, Université de Montreal, Montreal, 2021.
7. Cullinane, K., T. F. Wang, D. W. Song, and P. Ji. The Technical Efficiency of Container Ports: Comparing Data Envelopment Analysis and Stochastic Frontier Analysis. *Transportation Research Part A: Policy and Practice*, Vol. 40, No. 4, 2006, pp. 354–374.
8. Orive, A. C., J. I. P. Santiago, M. M. E. I. Corral, and N. González-Cancelas. Strategic Analysis of the Automation of Container Port Terminals Through BOT (Business Observation Tool). *Logistics*, Vol. 4, No. 1, 2020, p. 3.
9. Huynh, N., D. Smith, R. van Duin, M. Dulebenets, Y. Sun, P. Schonfeld, N. Hutson, et al. Challenges and the Road Ahead for Intermodal Freight Terminals. *Transportation Research Board Centennial Papers*, 2019. <https://onlinepubs.trb.org/onlinepubs/centennial/papers/AT050-Final.pdf>.
10. Donnelly, J. Qingdao Port Becomes the World's First Hydrogen-Powered and 5G Intelligent Port. *Port Technology* [Internet], June 23, 2022. <https://www.porttechnology.org/news/qingdao-port-becomes-the-worlds-first-hydrogen-powered-and-5g-intelligent-port/>. Accessed July 23, 2022.
11. Chen, X. An Empirical Analysis of Competitiveness of Yangshan Deep-Water Port: The Application of Intermodal Transportation and Automated Container Terminal. *Proc., 2nd International Workshop on Advances in Social Sciences (IWASS 2019)*, 2019, London, United Kingdom, pp. 111–122.
12. Garrido, J., S. Saurí, A. Marrero, Ü. Gül, and C. Rúa. Predicting the Future Capacity and Dimensions of Container Ships. *Transportation Research Record: Journal of the Transportation Research Board*, 2020. 2674: 177–190.
13. Mooney, T. Midsize Vessels Outpace Mega-Ships in Berth Productivity. *Journal of Commerce* [Internet], November 30, 2017. https://www.joc.com/port-news/data-show-berth-productivity-highest-midsize-ships_20171130.html. Accessed February 12, 2022.
14. Monteiro, M. F. *Productivity in the Container Port Business: Focus on the Mediterranean Range*. PhD thesis. Universiteit Antwerpen, Belgium, 2014.
15. Achterkamp, J. Improving Terminal Performance Mega-Ships Require Mega-Terminals. *Container Handling* [Internet], 2021. <https://wpassets.porttechnology.org/wp-content/uploads/2019/05/25182729/JOOST-PTIONLINE-5.pdf>. Accessed July 28, 2022.
16. Jobran, Y., and K. Gökhan. Examining the Efficiency of Automation in Container Terminals. *Journal of Transportation and Logistics*, Vol. 7, No. 1, 2022, pp. 137–155.
17. Burgos, L., P. Taneja, and M. van Koningsveld. Desktop Research into Historic Automation Projects of Brownfield Container Terminals. *Journal of Marine Science and Engineering*, Vol. 10, No. 5, 2022, p. 704.
18. Le-Griffin, H. D., and M. Murphy. Container Terminal Productivity: Experiences at the Ports of Los Angeles and Long Beach. *Proc., 1st I-NUF Conference*, 2006, pp. 1–21. <https://accessaintlaurentbeaupo.org/wp-content/uploads/2019/06/Le-Griffin-2006-Container-terminal-productivity-Experiences-of-ports-of-Los-Angeles-and-Long-Beach.pdf>
19. Saurí, S., and E. Martin. Space Allocating Strategies for Improving Import Yard Performance at Marine Terminals. *Transportation Research Part E: Logistics and Transportation Review*, Vol. 47, No. 6, 2011, pp. 1038–1057.
20. Nathan Associates Inc. *Principles of Monitoring Port Performance. Cap-Haitien Port Regulatory Strengthening Project*. USAID Office, Office of Infrastructure, Washington, D.C., 2018.
21. Wilmsmeier, G., B. Tovar, and R. J. Sanchez. The Evolution of Container Terminal Productivity and Efficiency Under Changing Economic Environments. *Research in Transportation Business & Management*, Vol. 8, 2013, pp. 50–66.
22. Lee, L. H., E. P. Chew, K. C. Tan, and Y. Wang. Vehicle Dispatching Algorithms for Container Transshipment Hubs. *OR Spectrum*, Vol. 32, No. 3, 2010, pp. 663–685.
23. Li, W., Y. Wu, M. E. Petering, M. Goh, and R. de Souza. Discrete Time Model and Algorithms for Container Yard Crane Scheduling. *European Journal of Operational Research*, Vol. 198, No. 1, 2009, pp. 165–172.

24. Chen, L., N. Bostel, P. Dejax, J. Cai, and L. Xi. A Tabu Search Algorithm for the Integrated Scheduling Problem of Container Handling Systems in a Maritime Terminal. *European Journal of Operational Research*, Vol. 181, No. 1, 2007, pp. 40–58.
25. Kim, K. H., and K. Y. Kim. Routing Straddle Carriers for the Loading Operation of Containers Using a Beam Search Algorithm. *Computers & Industrial Engineering*, Vol. 36, No. 1, 1999, pp. 109–136.
26. Yang, Y. C., and K. Y. Shen. Comparison of the Operating Performance of Automated and Traditional Container Terminals. *International Journal of Logistics Research and Applications*, Vol. 16, No. 2, 2013, pp. 158–173.
27. Fourgeaud, P. *Measuring Port Performance*. The World Bank, 2000, Washington, D.C., pp. 1–18.
28. Chu, F., S. Gailus, L. Liu, and L. Ni. The Future of Automated Ports. *McKinsey and Company*, 2018. <https://www.mckinsey.com/~media/McKinsey/Industries/Travel%20Transport%20and%20Logistics/Our%20Insights/The%20future%20of%20automated%20ports/The-future-of-automated-ports-final.pdf>.
29. Maritime Union of Australia. Inquiry into the Long-Term Productivity of Australia's Maritime Logistics System. *Australian Government Productivity Commission*, 2022. https://www.pc.gov.au/_data/assets/pdf_file/0004/336838/sub059-maritime-logistics.pdf.
30. Esmer, S. *Performance Measurements of Container Terminal Operations*. Maritime Business and Administration School, Dokuz Eylul University, Konak, Turkey, 2008.
31. Jordan, M. A. Quay Crane Productivity. *Proc., TOC Americas*, Miami, Liftech, 2002.
32. Mazloumi, M., and E. van Hassel. Improvement of Container Terminal Productivity with Knowledge about Future Transport Modes: A Theoretical Agent-Based Modelling Approach. *Sustainability*, Vol. 13, No. 17, 2021, p. 9702.
33. Voß, S., R. Stahlbock, and D. Steenken. A Survey on Operations Research Applications in Container Terminal Logistics. *Proc., Tristan V: The Fifth Triennial Symposium on Transportation Analysis*, Le Gosier, Guadeloupe, June 13–14, 2004.
34. Pjevecic, D., M. Nikolic, N. Vidic, and K. Vukadinovic. Data Envelopment Analysis of AGV Fleet Sizing at a Port Container Terminal. *International Journal of Production Research*, Vol. 55, No. 14, 2017, pp. 4021–4034. <https://doi.org/10.1080/00207543.2016.1241445>.
35. Lee, L. H., E. P. Chew, K. C. Tan, and Y. Wang. Vehicle Dispatching Algorithms for Container Transshipment Hubs. *OR Spectrum*, Vol. 32, No. 3, 2010, pp. 663–685.
36. Luo, J., and W. Yue. Scheduling of Container-Handling Equipment During the Loading Process at an Automated Container Terminal. *Computers & Industrial Engineering*, Vol. 149, 2020, p. 106848.
37. Humphreys, R. M., N. O. Biju, H. Tan, G. N. M. Ashley, D. Guillot, T. Mooney, A. Lane, and M. S. H. Wong. *The Container Port Performance Index 2021: A Comparable Assessment of Container Port Performance*. The World Bank, Washington, D.C., 2022.
38. GHD. Infrastructure Victoria: Second Container Port Advice: Estimated Capacity of the Port of Melbourne. 2017. https://www.infrastructurevictoria.com.au/wp-content/uploads/2019/04/GHD_Infrastructure_Victoria_second_container_port_advice_-_Estimated_capacity_of_the_Port_of_Melbourne_FINAL.pdf.
39. Xu, Y., and K. Ishiguro. Measuring the Efficiency of Automated Container Terminals in China and Korea. *Asian Transport Studies*, Vol. 5, No. 4, 2019, pp. 584–599.
40. Kugler, M., M. Brandenburg, and S. Limant. Automizing the Manual Link in Maritime Supply Chains? An Analysis of Twistlock Handling Automation in Container Terminals. *Maritime Transport Research*, Vol. 2, 2021, p. 100017.
41. Tanger Med Port Authority. Productivity Records at Tanger Med. *Safety4Sea*, October 1, 2021. <https://www.tangermedport.com/en/productivity-records-at-tanger-med/>. Accessed July 23, 2022.
42. Safety4Sea. APM Terminals Algeciras Sets New Port Moves Per Hour Record. *Safety4Sea*, April 3, 2019. <https://safety4sea.com/apm-terminals-algeciras-sets-new-port-moves-per-hour-record/>. Accessed July 23, 2022.
43. Camm, M. Container Volumes and Terminal Capacity in Northern Europe II. *Dyanmar White Paper*, Port Technology International, 2019. <https://www.porttechnology.org/wp-content/uploads/2019/05/Dynmar-2.pdf>.
44. Österreichische Verkehrszeitung. Euromax Terminal Rotterdam Set a New Record. *Österreichische Verkehrszeitung*, November 2, 2017. <https://oevz.com/en/euromax-terminal-rotterdam-set-a-new-record/>. Accessed July 13, 2022.
45. Wee, V. Hutchison's BEST Barcelona Terminal to Get 6 More Automatic Stacking Cranes. *Seatrade Maritime News*, May 23, 2018. <https://www.seatrade-maritime.com/europe/hutchison-s-best-barcelona-terminal-get-6-more-automatic-stacking-cranes>. Accessed July 24, 2022.
46. Raimund, D. *Impact of Automation on Port Container Terminals a Case Study on Transshipment Ports*. Erasmus School of Economics, Rotterdam, The Netherlands, 2020.
47. Aldama, Z. El puerto que sustituyó a los estibadores por robots y algoritmos. *El País*, October 28, 2019. https://elpais.com/retina/2019/10/28/innovacion/1572259447_860948.html#?ref=rss&format=simple&link=link. Accessed July 25, 2022.
48. WorldCargo News. Productivity and Remote Control STS Cranes. *WorldCargo News*, July 1, 2018. <https://www.worldcargonews.com/in-depth/in-depth/productivity-and-remote-control-sts-cranes>. Accessed July 23, 2022.
49. Cuppens, J. Terminal Automation: A Reality Check. *DP World*, 2018. https://www.tocevents-europe.com/content/dam/Informa/toc-events-europe/presentations/tech-toc-presentations/-automation-workshop/Jan_Cuppens.pdf. Accessed June 20, 2022.
50. Hadland, E., and R. Tomar. Ports & Terminals Insight Supplement. *Drewry*, 2022. https://www.drewry.co.uk/AcuCustom/Sitename/DAM/020/Example-Copy-PTI_06_2022_Monthly_Supplement.pdf. Accessed July 14, 2022.
51. Wiśnicki, B., L. Chybowski, and M. Czarnecki. Analysis of the Efficiency of Port Container Terminals with the Use of the Data Envelopment Analysis Method of Relative Productivity Evaluation. *Management Systems in Production Engineering*, Vol. 25, No. 1, 2017, pp. 9–15. <https://doi.org/10.1515/mspe-2017-0001>.

52. Rademaker, W. C. A. *Container Terminal Automation Feasibility of Terminal Automation for Mid-Sized Terminals*. Masters thesis. Delft University of Technology, Netherlands, 2007.
53. Hofmann, W., and F. Branding. Implementation of an IoT-and Cloud-Based Digital Twin for Real-Time Decision Support in Port Operations. *IFAC-PapersOnLine*, Vol. 52, No. 13, 2019, pp. 2104–2109.
54. Rao, S. K., and R. Prasad. Impact of 5G Technologies on Industry 4.0. *Wireless Personal Communications*, Vol. 100, No. 1, 2018, pp. 145–159.
55. Shadrin, S. S., O. Varlamov, and A. M. Ivanov. Experimental Autonomous Road Vehicle with Logical Artificial Intelligence. *Journal of Advanced Transportation*, Vol. 2017, 2017, pp. 1–10. <https://doi.org/10.1155/2017/2492765>.
56. Liu, R., Y. Li, L. Tao, D. Liang, and H. T. Zheng. Are we Ready for a New Paradigm Shift? A Survey on Visual Deep MLP. *Patterns*, Vol. 3, No. 7, 2022, p. 100520. <https://doi.org/10.1016/j.patter.2022.100520>.
57. Sun, T. X., X. Y. Liu, X. P. Qiu, and X. J. Huang. Paradigm Shift in Natural Language Processing. *Machine Intelligence Research*, Vol. 19, No. 3, 2022, pp. 169–183.
58. Nain, G., K. K. Pattanaik, and G. K. Sharma. Towards Edge Computing in Intelligent Manufacturing: Past, Present and Future. *Journal of Manufacturing Systems*, Vol. 62, 2022, pp. 588–611. <https://doi.org/10.1016/j.jmsy.2022.01.010>.